

Tilak Patel

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TECHNICAL SKILLS

Languages: Python, C/C++, Java, C#/.NET, SQL, Assembly (x86), JavaScript

Frameworks: FastAPI, Spring Boot, React, Node.js, PyTorch, LangGraph, Qt/PySide6

Tools & Platforms: AWS (Redshift, Bedrock), Docker, Git, Linux, GitHub Actions, MongoDB, Grafana, MCP

EXPERIENCE

Amazon Web Services (AWS)

Herndon, VA

Technical Infrastructure Program Manager Intern

Sept 2026 – Present

- Developing data center planning tools modeling server placement constraints for generative AI capacity
- Architecting Python multi-agent pipeline querying Redshift via MCP to flag stale capacity data and notify owners
- Scoping automation requirements with delivery stakeholders by mapping manual blocker tracking workflows

RTX Corporation

Hartford, CT

Application Transformation Intern

June 2026 – Aug 2026

- Helped design the application modernization value stream and roadmap targeting 2028 enterprise goals
- Assessed enterprise applications across discovery and execution phases, migrating workloads to cloud and on-prem
- Collaborated with application teams to plan and execute transitions to RTX's Xeta Cloud modernization platform

Bose Corporation

Framingham, MA

Software Engineer Co-op

Jan 2026 – June 2026

- Built full-stack log analysis platform with 60+ REST endpoints, streaming, and session persistence using FastAPI
- Designed 3-phase processing pipeline with 25+ custom tools, reducing firmware debugging from hours to minutes
- Shipped cross-platform desktop app (React, Electron, Copilot SDK) integrating Jira, Confluence, and Artifactory

Pendar Technologies

Boston, MA

Laser Software Engineer Co-op

July 2025 – Dec 2025

- Built real-time data acquisition app in Qt/PySide6 for laser sensor testing, fully replacing legacy LabVIEW system
- Implemented automated overnight testing with CSV export, eliminating hours of manual data collection daily

SRC, Inc.

Syracuse, NY

Machine Learning Engineer Intern

April 2025 – July 2025

- Built pipeline extracting knowledge triplets from 500+ radar docs at 90% accuracy using Pydantic and Docling
- Fed extracted triplets into knowledge graphs via LangGraph, automating analysis and cutting review time by 70%

PROJECTS/RESEARCH

Research – GPU Checkpoint-Restart (Prof. Gene Cooperman) | C, MPI, NCCL, Linux

- Profiled MPI checkpoint-restart overhead in MANA using flamegraphs, identifying communication bottlenecks
- Implementing GPU checkpoint-restart for NCCL collectives, enabling fault recovery in multi-GPU training jobs

Game Boy Emulator | C++, SDL3, CMake

[GitHub](#)

- Developed Game Boy (DMG) emulator passing Blargg's CPU test suite, implementing all 500+ LR35902 opcodes
- Implemented scanline-based PPU and memory-mapped I/O via SDL3, running Game Boy ROMs at 60 FPS

Devspace – HackBeanpot 1st Place | Java, React, Spring Boot, Docker, CUDA, WebSockets

[GitHub](#)

- Built collaborative cloud IDE with GPU-accelerated Docker execution (CUDA/C++/Python) on Jetson Nano
- Implemented real-time multi-user editing with cursor sync and conflict resolution via WebSockets

Open Source – github/awesome-copilot (29K+ stars) | Python, Copilot SDK

[GitHub](#)

- Wrote error-recovery hooks and PyInstaller frozen-build recipes for the Copilot SDK, merged into official repo

EDUCATION

Northeastern University

Boston, MA

B.S. Computer Science | GPA: 3.6/4.0

May 2027

Coursework: Grad AI, Grad HPC, Algorithms, Computer Systems, Systems Security, Robotics, Theory of Computation